

OOP Lab Tasks

Class & Object in C++

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Task 1: Basic Class & Object

Create Class: Student

- ▶ Data Members:
 - ▶ Name
 - ▶ Roll Number
 - ▶ Marks
- ▶ Use public access specifier
- ▶ Create object in main()
- ▶ Take input from user
- ▶ Display student information

Task 2: Multiple Objects

Create Class: Book

- ▶ Attributes:
 - ▶ Book Title
 - ▶ Author Name
 - ▶ Price
- ▶ Create 3 objects
- ▶ Take input for all
- ▶ Display details

Task 3: Method Inside Class

Create Class: Rectangle

- ▶ Data Members:
 - ▶ Length
 - ▶ Width
- ▶ Method:
 - ▶ calculateArea()
- ▶ Create object, input values, display area

Task 4: Result Management System

Create Class: Result

- ▶ Data Members:
 - ▶ Name
 - ▶ Roll No
 - ▶ 5 Subject Marks
- ▶ Methods:
 - ▶ inputData()
 - ▶ calculateAverage()
 - ▶ displayResult()

Task 4: Requirements

- ▶ Create 2 student objects
- ▶ Validate marks (0–100)
- ▶ Display:
 - ▶ All marks
 - ▶ Average
 - ▶ Pass/Fail (Pass 50)

Task 5: Employee Salary System

Create Class: **Employee**

- ▶ Data Members:
 - ▶ Name
 - ▶ Employee ID
 - ▶ Basic Salary
- ▶ Methods:
 - ▶ inputData()
 - ▶ calculateBonus()
 - ▶ displayDetails()
- ▶ Create three files:
 - ▶ class_header.h
 - ▶ class_header.cpp
 - ▶ main.cpp
- ▶ Define methods using scope resolution (::)
- ▶ Create multiple objects
- ▶ Display output for:
 - ▶ Salary
 - ▶ Bonus
 - ▶ Final Salary

- ▶ Learned Class and Object fundamentals
- ▶ Created single and multiple objects
- ▶ Implemented methods inside class
- ▶ Built real-world systems
- ▶ Practiced modular programming with multiple files